

A First-Principles Derivation of the Electroweak Mixing Angle and Z-Boson Resonance Correction

ABSTRACT

This paper presents a parameter-free, first-principles derivation of the pure background weak mixing angle ($\sin^2\theta_W$) and resolves the historical discrepancy between theoretical calculations and direct SI laboratory measurements at the Z-pole. By analysing the structural geometry of a five-dimensional, tension-bearing discrete substrate, the unloaded electroweak coupling is derived strictly from geometric projection and boundary potential losses:

$$\sin^2\theta_{W_pure} = (\Delta_s / (\beta_s \Xi)) (1 - \delta_{weak}) (1 - \Sigma J) = 0.231236$$

In terrestrial laboratory frames, the massive, rotating, spin-1 vector boson of the Z-pole projects its wave envelope across all four coordinate quadrants of our four-dimensional timeline (R_4). Under the Hysteresis Partition Principle, this projects a cumulative environmental coordinate tax of exactly four units of the fundamental background response delay:

$$4\mathbb{H} = 0.000306$$

Adding this coordinate tax to the pure background coupling angle yields the terrestrial effective weak mixing angle:

$$\sin^2\theta_{eff_terrestrial} = \sin^2\theta_{W_pure} + 4\mathbb{H} = 0.231542$$

This derived value matches the raw SI laboratory measurement of the effective leptonic weak mixing angle at the Z-pole (0.23153 ± 0.00016) with absolute precision, explaining the experimental discrepancy as a direct consequence of testing a four-quadrant bosonic projection in a dynamically loaded vacuum.

1. INTRODUCTION

The weak mixing angle, or Weinberg angle (θ_W), is a cornerstone parameter of the standard electroweak theory, characterising the relative coupling strengths of the electromagnetic and weak interactions. Despite its empirical success, the standard model

of particle physics is unable to derive θ_W from first principles, treating it instead as an arbitrary, experimentally determined input. Furthermore, high-precision electroweak measurements at the Z-pole reveal a persistent, highly scrutinised offset between standard theoretical calculations and the direct effective leptonic weak mixing angle measured in laboratories.

This paper addresses both challenges by deriving the weak mixing angle from the geometric projection limits of a discrete, five-dimensional information-bearing substrate. It is critical to note at the outset that the scope of this paper is strictly limited to the electroweak sector and the precise derivation of the weak mixing angle at the Z-pole. The reader is referred to the foundational masterwork for the comprehensive, first-principles derivation of the substrate's global invariants — including the Prime Seed ($\dagger = 6.2963456314$), the five-dimensional coordinate projection factor, the baseline lattice spacing, and the complete unified Lagrangian.

Assuming these foundational parameters as established background constants, this paper focuses exclusively on demonstrating how the electroweak mixing angle and its terrestrial laboratory offset emerge naturally from the geometry of the projection path and the localised hysteresis delay of the loaded vacuum.

2. GEOMETRIC FOUNDATIONS AND STARTING PRIMITIVES

All derivations in this work are conducted within a normalised, dimensionless natural unit system where the fundamental yield limit of the substrate is set to unity ($\mathfrak{m} = 1$). The physical parameters of the electroweak sector are governed by the following structural constants:

- The Prime Seed ($\dagger$): $\dagger = 6.2963456314$. This defines the fundamental tension-to-yield ratio of the substrate, setting the Compton–Schwarzschild scale of the lattice.
- The Normalised Yield Limit ($\mathfrak{m}$): $\mathfrak{m} = 1$. This establishes the boundary of matter formation, where continuous superposed density condenses into stable coordinate-locked configurations.
- The EM-Sector Projection Factor (Δ_s): $\Delta_s = \dagger / \pi^2 = 0.637851$. This represents the geometric ratio required to map unitless substrate magnitudes into four-dimensional electromagnetic observables.
- The EM Brook Constant (β_s): $\beta_s = \mathfrak{m}_s \beta = 1.2810675$. This parameter describes how tension spreads and saturates across the available propagation routes of the active electromagnetic channel.

- The Bars/Recursion Base (Ξ): $\Xi = 2$. This is the fundamental doubling factor for structural projection across the stacked organisational layers.
- The Weak-Sector Projection Loss (δ_{weak}): $\delta_{\text{weak}} = 1 / \sqrt{2} - \Delta_s = 0.069256$. This represents the residual projection offset of the weak sector.
- The Junction Gap ($\mathcal{J}$): $\mathcal{J} = (\dagger / 2\pi - \eta) - \mathcal{H} = 0.0019796024$. This defines the non-zero structural residue between the substrate's total cyclic capacity potential ($\Sigma J = \dagger / 2\pi - \eta$) and the non-expressible response delay paid via coordinate-boundary hysteresis ($\mathcal{H}$).
- The Hysteresis Constant ($\mathcal{H}$): $\mathcal{H} = 1 / (2\pi \dagger) = 0.000076456$. This is the non-expressible response delay paid during coordinate-boundary transitions.

3. FIRST-PRINCIPLES DERIVATION OF THE PURE ELECTROWEAK MIXING ANGLE

In the substrate framework, physical interactions are not abstract forces mediated by virtual particles in empty space; instead, they represent the propagation of stress-energy patterns across nested, recursive layers of the five-dimensional lattice. The organisation of these layers is governed by the fundamental dimensional recursion relation:

$$R_n = \{ S \mid S \subseteq R_{n-1} \}$$

Where each higher-dimensional layer R_n consists of all possible arrangements and subsets of the states immediately below it (R_{n-1}), building up structural complexity step-by-step from the fundamental substrate bit R_0 to the R_5 superposed multiverse.

Electroweak Projection Paths and Recursion Symmetries

Under this dimensional recursion hierarchy, the weak and electromagnetic interactions correspond to distinct, geometrically restricted pathways through the nested layers:

The Electromagnetic Path ($R_2 \rightarrow R_3 \rightarrow R_4$): Transverse tension oscillations (photons) propagate directly across the 2D sheets of the R_2 tension plane, intersect to form 3D spatial frames in R_3 , and project into the classical, resolved macroscopic R_4 timeline. Because this path is direct and remains entirely within the resolved, classical domain, it carries no Junction Gap coordinate slip and incurs no environmental hysteresis delay ($\Sigma J = 0$).

The Weak Path ($R_2 \rightarrow R_3 \rightarrow R_4 \rightarrow R_5 \rightarrow R_4$): Weak decay and threshold transitions require the stress-energy pattern to cycle through the superposed, five-dimensional multiverse layer

(R_5) before returning to the classical timeline. This complete, five-stage closed loop experiences two distinct physical boundaries:

- One Yield-Boundary Slip (J): Paid when crossing the matter-formation boundary to enter the superposed R_5 domain, representing the mechanical gear slip between cyclic tension and the intrinsic yield limit.
- Two Hysteresis delays ($2H$): Paid on the R_5 return path, representing the mandatory response delay required for the un-collapsed tension paths of the 5D superposed layer to realign and lock back onto the macroscopic coordinate axes.

The weak mixing angle, $\sin^2\theta_W$, arises from the structural balance between this projection efficiency, the dimensional recursion scaling, and the boundary losses at the yield threshold. Mathematically, it is given by the structural relation:

$$\sin^2\theta_{W_pure} = (\Delta_s / (\beta_s \Xi)) (1 - \delta_{weak}) (1 - \Sigma J)$$

Where the term $(1 - \Sigma J)$ represents the slip-behaviour and hysteresis boundary correction. The total boundary potential available per loop (ΣJ) is the combined hardware load of the Junction Gap and the Hysteresis response delay:

$$\Sigma J = J + H = 0.0019796024 + 0.000076456 = 0.002056056$$

This single-loop potential (ΣJ) represents the raw boundary loss of the transition. Evaluating the terms of the structural relation yields:

- The raw projection ratio: $\Delta_s / (\beta_s \Xi) = 0.637851 / (1.2810675 \times 2) = 0.248954$
- The weak-sector loss correction: $1 - \delta_{weak} = 1 - 0.069256 = 0.930744$
- The boundary potential correction: $1 - \Sigma J = 1 - 0.002056056 = 0.997944$

Evaluating the product of these three terms:

$$\sin^2\theta_{W_pure} = 0.248954 \times 0.930744 \times 0.997944 = 0.231236$$

The value of 0.231236 represents the pure, unloaded background parameter of the electroweak mixing angle in a frictionless vacuum, free from the coordinate loading of a dynamic, active environment.

4. THE HYSTERESIS PARTITION AND THE Z-POLE COORDINATE TAX

The pure background value of $\sin^2\theta_{W_pure} = 0.231236$ represents the theoretical limit in an unperturbed reference frame. However, physical laboratory measurements are conducted within un-subtracted terrestrial frames loaded by the active, sub-yield background of our local environment.

The cumulative hysteresis loading experienced by a physical particle is determined strictly by the number of independent, intersecting tension junctions required to maintain axial closure in its corresponding recursion layer. This topological rule establishes the Singlet and Triplet Hysteresis Partition.

The Z-boson is a massive spin-1 vector boson whose rotating wave envelope projects across all four orthogonal coordinate quadrants of our four-dimensional timeline (R_4). Under the Hysteresis Partition Principle, the active axes of this massive, rotating projection experience a cumulative environmental coordinate tax of exactly four units of the fundamental background response delay:

$$\text{Z-pole Hysteresis Tax} = 4\mathbb{H} = 4 \times 0.000076456 = 0.000305824 \approx 0.000306$$

This coordinate tax acts as a mandatory propagation delay applied to the bosonic wave envelope as it cycles through the quadrants of the physical frame. Adding this environmental coordinate tax to the unloaded background coupling parameter yields the effective terrestrial weak mixing angle measured in laboratories:

$$\sin^2\theta_{\text{eff_terrestrial}} = \sin^2\theta_{W_pure} + 4\mathbb{H}$$

$$\sin^2\theta_{\text{eff_terrestrial}} = 0.231236 + 0.000306 = 0.231542$$

5. EXPERIMENTAL VALIDATION AND CODATA COMPARISON

The derived terrestrial effective value of 0.231542 stands in stark contrast to the underivable parameters of the Standard Model, matching high-precision empirical data with absolute precision. The international baseline reference values for the effective leptonic weak mixing angle measured directly at the Z-boson pole yield an experimental average of:

$$\sin^2\theta_{\text{eff}}^{\text{lept}}(M_Z) = 0.23153 \pm 0.00016$$

The Mechaniverse derived output of 0.231542 falls perfectly within the centre of the experimental error bars. This absolute mathematical alignment demonstrates that the

long-standing discrepancy between baseline vacuum couplings and laboratory measurements is not an indication of missing particles or free parameters, but is the predictable geometric consequence of testing a four-quadrant bosonic projection within a dynamically loaded terrestrial vacuum.

6. CONCLUSION

This paper has demonstrated that the electroweak mixing angle is an emergent parameter governed strictly by the structural geometry and boundary potential losses of a discrete 5D substrate. By evaluating the recursion path scaling, the model derives the unloaded background coupling $\sin^2\theta_{W_pure} = 0.231236$ entirely from first principles. Furthermore, by accounting for the environmental coordinate tax incurred across the four quadrants of the R_4 timeline, the terrestrial effective angle is derived as $\sin^2\theta_{eff_terrestrial} = 0.231542$, matching high-precision laboratory measurements at the Z-pole.

The scope of this paper is restricted to core derivations and key results. Full mathematical derivations, extended explanations, supporting numerical calculations, and complete nomenclature are provided in the main Mechaniverse working paper [0].

REFERENCES

- [0] Fenn, J. (2026). "The Mechaniverse: A Unified Information-Substrate Model of the Universe," Zenodo, Working Paper. doi: 10.5281/zenodo.18722388
- [1] Glashow, S. L. (1961). Partial-symmetries of weak interactions. *Nuclear Physics*, 22(4), 579–588.
- [2] Weinberg, S. (1967). A model of leptons. *Physical Review Letters*, 19(21), 1264–1266.
- [3] Salam, A. (1968). Weak and electromagnetic interactions. *Nobel Symposium*, 367.
- [4] Mohr, P. J., Newell, D. B., & Taylor, B. N. (2016). CODATA recommended values of the fundamental physical constants. *Reviews of Modern Physics*, 88(3), 035009.
- [5] Blaszyński, J. (2024). The Dynamic Space-Time Metric (DSM): Foundations of Elastic Vacuum Dynamics. *Journal of Mathematical Physics*, 65(4), 042501.
- [6] Particle Data Group. (2024). Review of particle physics. *Progress of Theoretical and Experimental Physics*, 2024(8), 083C01.
- [7] 't Hooft, G. (1993). Dimensional reduction in quantum gravity. *arXiv:gr-qc/9310026*.

[8] Verlinde, E. P. (2011). On the origin of gravity and the laws of Newton. *Journal of High Energy Physics*, 2011, 29.

[9] Jacobson, T. (1995). Thermodynamics of spacetime: The Einstein equation of state. *Physical Review Letters*, 75(7), 1260–1263.

[10] LHCb Collaboration. (2022). Test of lepton flavour universality in B-meson decays at LHCb. CERN, CERN-EP-2022-250.